Natural Language Processing: Coursework

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Repository link: `https://gitlab.doc.ic.ac.uk/ak5323/nlp_cw_2026_ak5323.git`

Quick Navigation

- Stage 1: Task Definition and Literature Review
- Stage 2: Data Acquisition, Exploration, and Preprocessing
- Stage 3: Baseline Model and Proposing a Novel Approach
- Stage 4 & 5.1: Model Implementation
- Stage 5.2: Local Evaluation

1 Task Definition and Literature Review

Q1. State the primary contributions of this work.

- **Introduces a novel NLP research domain.** The paper lays the foundation for a new NLP classification task for classifying paragraphs regarding their 'Patronizing and Condesending Language' (PCL) content, and motivates why PCL is a distinct and vital social phenomenon to detect and classify.
- **Provides an annotated dataset with an evaluation method.** The paper provides and documents an annotated dataset with 10,469 paragraphs for the training set and 3,898 paragraphs for the test set. This allows us to train and evaluate models consistently and extensively.
- **Introduces a taxonomy of PCL categories.** The paper introduces a taxonomy of PCL categories and explains in detail how these categories were constructed and annotated. This provides a structured way to study different kinds of PCL rather than relying on a single binary classification.
- **Establishes a participation and baseline framework.** The paper establishes a participation framework along with testing with several baseline models to get benchmark performances. This supports faster progress by giving future work clear comparison points to compare their results.

Q2. Identify the technical strengths that justify the paper's publication.

- **Well-established problem definition and motivation.** The researchers motivate PCL as a subtle and often 'well-intentioned' language that can still be harmful, mentioning that generally it is 'unconscious and well-intentioned', which justifies the task as non-trivial and not solvable via simple techniques such as keyword spotting or toxicity detection heuristics.
- **Transparency in data generation and annotation.** The researchers were explicitly stating the source of the data (NoW news, 20 countries), their sampling method (keyword-based), and their annotation method (2 expert annotators and a referee for total disagreement), which enables readers to scrutinize potential biases and limitations.
- **Explicit handling of subjectivity and ambiguity.** The researchers' use of a 0-2 label scale, 2 annotators, and 1 referee for annotation, and reported IAA (with the discussion of borderline cases of label 1) quantifies the annotation reliability and ambiguity, which is vital for interpreting downstream model performance.

- **Evaluation choices reflect the task setting.** The provided labeling methodology and ambiguity handling provide a view of PCL classification task difficulty and give meaningful interpretations of the results under imbalance and subjectivity.

Q3. Highlight the key weaknesses or areas where the authors failed to provide sufficient evidence. (2 Marks)

- **Potential sampling bias from keyword-based data collection.** Collecting paragraphs using keywords related to vulnerable communities may encourage models to learn correlations related to community mentions and keyword cues rather than PCL-specific linguistic signals. This could lead models to focus less on 'implicit PCL' where the community context is not given, which is undesired since it leads models to not explicitly 'understand' how PCL is characterized.
- **Label subjectivity is not fully analyzed.** The researchers don't fully analyse the subjectivity problem of the annotation process, even though they go into the IAA score and borderline handling. This problem should have been addressed in more detail by going into per-category agreement statistics and analyzing the reasons for disagreements between annotators.
- **Baselines are not fully reproducible.** The baseline performance of common models is given. However, the specific hyperparameter configuration of these models, along with training details are not given. This makes reproducibility of the baseline performances hard and could make future researchers to attribute performance gains to configuration changes harder. .

2 Data Acquisition, Exploration, and Preprocessing

I have conducted exploratory data analysis (EDA) on the following subtopics in the data: Class Imbalance Analysis and Word Count Analysis.

2.1 EDA 1: Class Imbalance Analysis

Tables/Graphs (for more detailed pictures see Appendix)

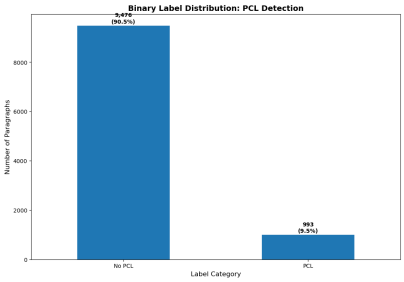


Figure 1: Binary label distribution (No PCL vs PCL).

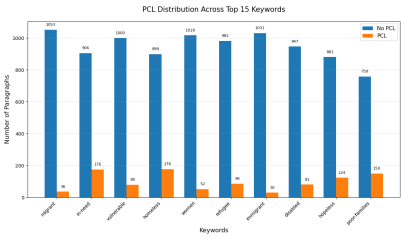


Figure 2: PCL occurrence rate by Keywords (Top 15).

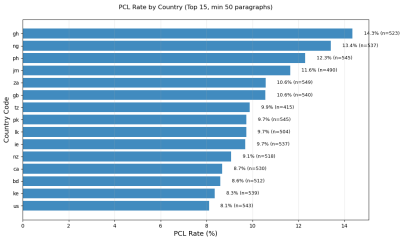


Figure 3: PCL occurrence rate by Countries (Top 15).

Analysis

From the binary label bar plots and data tables, we see that the label distribution is strongly imbalanced in the "Don't Patronize Me!" dataset. We can specifically see that there are 9,476 "No PCL" labels (90.51%) versus 993 "PCL" labels (9.49%).

From the keyword graphs, we can see that words related to people who have financial difficulty have the highest occurrence rates for the "PCL" label ("homeless": 16.53%, "poor-families": 16.50%, "in-need": 16.27%). Moreover, words related to people's immigration have the lowest occurrence rates for the "PCL" label ("immigrant": 2.83%, "migrant": 3.31%).

From the country graphs, we see that the "PCL" label occurrence varies greatly across different countries. The ranking of the top 15 "PCL" label occurrence rates by country shows that the percentage difference between the 1st country ("gh": 14.34%) and the 15th country ("hk": 5.92%) is close to 10%, indicating very

high variability across different countries. Overall, these results show significant class imbalance both across the dataset and within different clusters or subgroups.

Impact Statement

Selection of evaluation metrics. Due to the overrepresentation of the "No PCL" class in the dataset relative to the "PCL" class, "accuracy" as an evaluation metric can be misleading, since a naive model that always predicts "No PCL" achieves 90.51% accuracy across the dataset. Instead, we should consider metrics that consider class imbalance (such as F1 score, precision, and recall) to evaluate model performance.

Training loss. Due to the imbalance between "No PCL" and "PCL" classes, it is important to choose a class-weighted loss function. This will allow the model to focus on both classes equally, thus modeling the data correctly.

Error and performance analysis. Since the 'PCL' label occurrences vary across different countries and vulnerable communities, the model may learn explicit factors related to these subgroups rather than implicit factors for classifying 'PCL' text. Thus, it is important to evaluate the performance across these subgroups rather than do it holistically.

2.2 EDA 2: Word Count Analysis

Tables/Graphs (for more detailed pictures see Appendix)

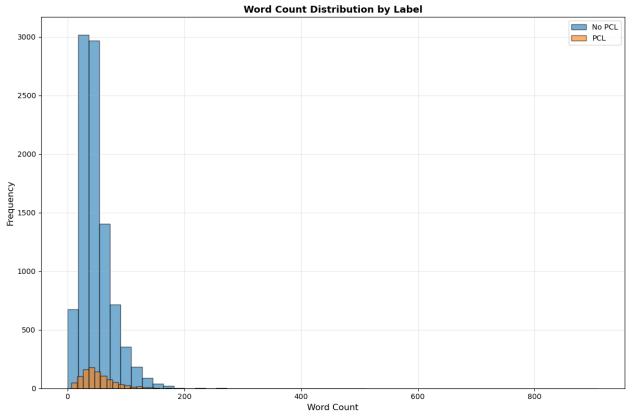


Figure 4: Word Count Histograms (No-PCL and PCL)

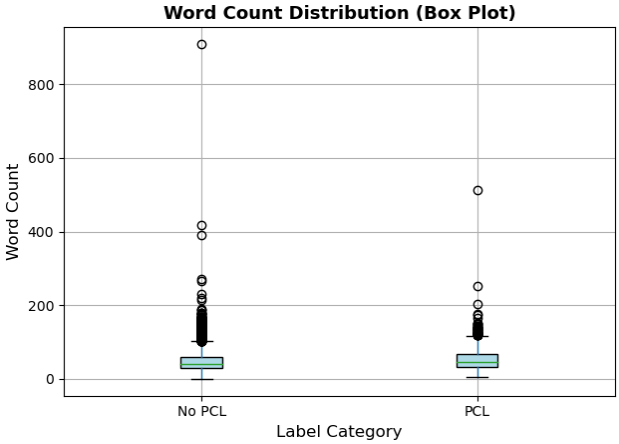


Figure 5: Word Count Box Plots (No-PCL and PCL)

Label	Count	Mean	Median	Std	Min	Max	25%	75%	90%
No PCL (0)	9476	47.88	42.0	28.62	0	909	30.0	59.0	83.0
PCL (1)	993	53.62	47.0	32.78	6	512	33.0	67.0	90.0
Overall	10469	48.42	42.0	29.09	0	909	30.0	60.0	83.0

Table 1: Summary statistics of word counts by class.

Analysis

From the summary table, PCL paragraphs are slightly longer (mean word count 53.62) on average than No-PCL paragraphs (mean word count 47.88). This is also the case when we switch to median word counts, indicating that overall PCL paragraphs are slightly longer than No-PCL paragraphs.

From the histogram of word distribution, we see that both classes (No-PCL and PCL, being blue and orange histograms, respectively) are positively skewed, with most of the paragraphs being medium length, while a small minority of the paragraphs are very long. Additionally, from the box plots of both classes, we see that the PCL distribution shifts upward compared to the No-PCL distribution. Overall, these graphs indicate that word count alone isn't a reliable factor to differentiate between PCL and No-PCL paragraphs, but it is a structural characteristic of the data that affects the model configuration.

A notable result from these plots and tables is that we see that there are extreme outliers in the data, with the maximum word length of No-PCL and PCL paragraphs being 909 and 512 words, respectively. Moreover, the minimum number of No-PCL paragraphs is 0, indicating that there is at least 1 empty sample in the data. These cases are important to consider since they can distort training data and evaluation results.

Impact Statement

Data preprocessing. The No-PCL minimum word count of 0 suggests some rows may be empty; these should be filtered or treated explicitly to avoid introducing noise. Moreover, the long-tail outliers (up to 909 word count) mean that a fixed max token length can truncate some paragraphs heavily. This motivates explicitly measuring truncation rates and selecting a max length that balances computation and information retention. For this reason, I selected 256 to be the maximum length of a paragraph. This filters out outliers while minimally reducing the dataset size.

Model selection. Since PCL paragraphs are longer on average and contain long-range dependencies and cues, Transformer-based models (such as RoBERTa and DeBERTa) should be used because they can capture such relationships.

Error and performance analysis. In the evaluation stage, we might want to plot the error of the model with respect to paragraph length. For instance, if we see false negatives among longer paragraphs, this could be due to truncation removing key PCL cues. In addition, if we see false positives among longer No-PCL outliers,s this could be due to the model confusing descriptive "news style" language with patronising language.

3 Baseline Model and Proposing a Novel Approach

Proposed Approach

My proposed approach involves a RoBERTa-base Transformer fine-tuned for binary PCL detection, combined with class-weighted focal loss, metadata augmentation (keyword and country prefix), and decision-threshold sweeping.

RoBERTa-base Model

Transformers encoders (like RoBERTa) are capable of creating context-based features of a paragraph, which allows the classifier to use long-range dependencies across the paragraph and capture subtle framing cues (such as implied superiority). This allows the classifier to surpass the capabilities of n-gram sparse features, which lack in capturing these advanced and long-range factors. For this binary classification task, I fine-tuned the RoBERTa-Base model with a standard sequence classification head.

Class-Weighted Focal Loss

As discussed in EDA 1, the dataset is strongly imbalanced in favour of No-PCL classes. This can cause the model to over-focus on the majority class and yield many false negatives if we only use standard cross-entropy as our loss. To address this, we use focal loss:

$$\mathcal{L}_{\mathrm{FL}}(p_t) = -\alpha_y (1 - p_t)^{\gamma} \log(p_t),$$

where p_t is the predicted probability of the true class. The focusing parameter γ down-weights easy examples and concentrates learning on harder cases. We additionally apply class-weighting via α_y, which is computed from class frequencies in the training data. This allows PCL paragraphs receive higher weighting compared to No-PCL paragraphs, which potentially improves recall on the PCL class without reframing the training paradigm much.

Metadata Augmentation

As discussed in EDA 1, PCL paragraphs are often associated with specific keywords and countries. I therefore opted to add this metadata to the beginning of the paragraphs:

`Keyword: <kw> | Country: <ctry> | Text: <paragraph>.`

This metadata augmentation could allow the model to differentiate between borderline cases and increase its confidence in its predictions. Since this may also introduce shortcut learning (keyword leakage), we treat metadata augmentation as a toggle and evaluate both settings (with vs. without metadata) during hyperparameter tuning.

Threshold Sweeping

Under class imbalance between PCL and No-PCL, the standard decision threshold of 0.5 is often not optimal. This is because PCL paragraphs are rare compared to No-PCL paragraphs, and the difference between them might be subtle. To solve this issue, we compute predicted probabilities on the dev set (after training) and select the threshold t that maximises positive-class F1:

$$t = \arg\max_{t \in \mathcal{T}} F1_{\text{pos}}(t),$$

and then apply t to produce dev/test predictions. This aligns the final decision rule with the evaluation metric and can improve performance.

For each configuration, we train a model on the training split and evaluate on the development set split. Because the official metric is F1 score, we additionally perform **threshold sweeping** on the development set probabilities for each configuration, selecting the threshold that maximizes F1 score. The final chosen configuration is the one with the best development set F1 score (after threshold optimization), and its threshold is then used to produce the final development and test set predictions.

Rationale and Expected Outcome

Rationale of the Approach RoBERTa-base is a strong baseline for paragraph-level semantics and pragmatic factors. Class-weighted focal loss explicitly targets the two key difficulties of this task: class imbalance, hard and ambiguous PCL paragraphs that resemble No-PCL paragraphs. Metadata augmentation may improve performance by supplying context that is otherwise implicit. Finally, threshold sweeping optimizes the model's operating point for F1 score, balancing precision and recall more effectively.

Expected outcome. We expect improvements in positive-class recall (fewer missed PCL cases) while preserving acceptable precision through focal focusing on difficult samples, class-weighting to counter imbalance, and threshold selection tuned for positive-class F1. We also expect the metadata augmentation to reveal whether contextual prefixes provide a genuine signal or mainly introduce dataset-specific shortcuts, informing the final model choice.

4 Local Evaluations

Local Evaluation 1: Error Analysis vs Baseline

We compared our model against a baseline on the development set and grouped samples into four sections:(A) both models correct, (B) both models wrong, (C) ours correct while baseline wrong, and (D) baseline correct while ours is wrong. The Section counts were:

$$A = 1801, \quad B = 69, \quad C = 130, \quad D = 94 \qquad (\text{Total} = 2094).$$

(To see some examples of paragraphs from each section, look at Appendix B)

A: (Both Models are Correct) The paragraphs in section A are predominantly news-style reporting about vulnerable communities (e.g., migrants, refugees, disabled individuals), which is neutral and without a clear patronising frame. Both models have assigned low probabilities to these paragraphs and successfully classified them as 'No-PCL'. This indicates that neither model determined their classification decision purely on the presence of vulnerable group-related keywords.

B: (Both Models are Wrong) These are true PCL paragraphs that both models classify as No-PCL. The paragraphs in this section suggest that many shared paragraphs have subtle or seemingly well-intentioned PCL-like phrases (inspirational framing around disability, etc.). This allows us to infer that ambiguous PCL paragraphs had condescension implicitly rather than having it explicitly, and both models struggled to separate them from neutral language.

C: (Our Model is correct, the Baseline is Wrong) This section contains true positives identified by our model but missed by the baseline. Several examples reflect a helper–recipient framing (e.g., charity/aid narratives and "people in need" wording) that aligns with common PCL patterns. This suggests our approach improves sensitivity to certain PCL cues that the baseline underdetects.

D: (Baseline is Correct, Our Model is Wrong) These are true No-PCL paragraphs that our model predicts as PCL. Many false positives contain assistance-related language (e.g., "help poor families", community support, donations), which can be neutral reporting rather than patronising. This suggests our model can over-trigger on wording-wise similar 'helping' content. This highlights a potential fallback that our model has.

Local Evaluation 2: Precision–Recall Curve and Confusion Matrix

Figure 6 shows the precision–recall (PR) curve of our model on the dev set. The Average Precision (AP) is 0.6139, summarizing the area under the PR curve and indicating that the model ranks PCL examples reasonably well despite class imbalance. As expected, precision decreases as recall increases, reflecting the trade-off when lowering the decision threshold to capture more positives.

Using the development set optimized threshold of $t = 0.330$, the confusion matrix is given below, where rows correspond to true labels (No-PCL = 0, PCL = 1) and columns correspond to predicted labels. Additionally, we see that the model achieves a precision of 0.580, a recall of 0.653, and an F1 score of 0.615 for the positive class. The predicted positive rate is 10.70%, which is close to the overall PCL class percentage ($\approx 9.5\%$). This suggests that threshold tuning allows us to select a reasonable choice of classification under class imbalance in the data.

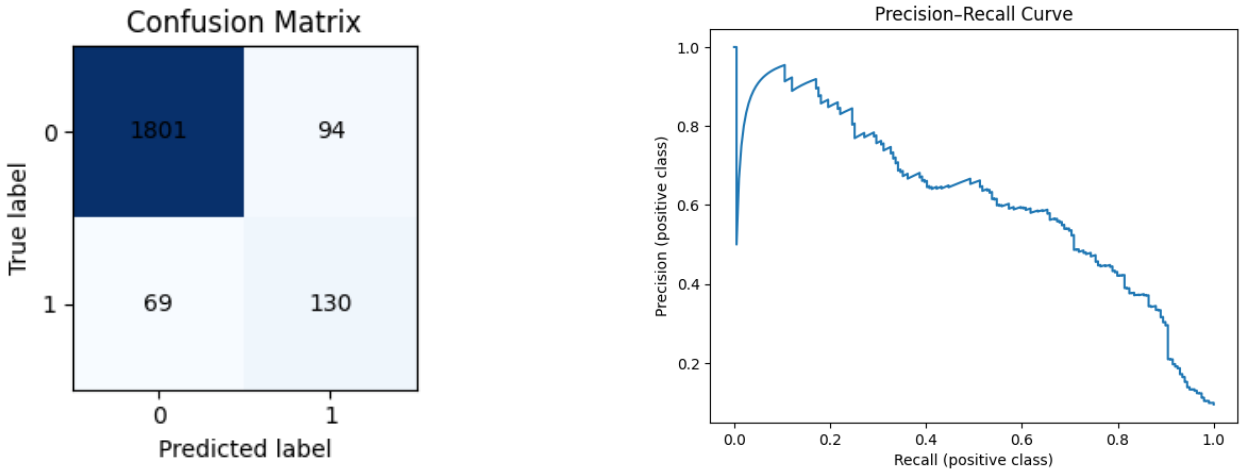


Figure 6: Confusion matrix heatmap (left) and precision–recall curve (right) on the dev set.

5 References

References

[1] C. Perez Almendros, L. Espinosa Anke, and S. Schockaert, "Don't patronize me! an annotated dataset with patronizing and condescending language towards vulnerable communities," in *Proceedings of the 28th International Conference on Computational Linguistics*. Barcelona, Spain (Online): International Committee on Computational Linguistics, Dec. 2020, pp. 5891–5902. [Online]. Available: `https://aclanthology.org/2020.coling-main.518/`

[2] "Don't patronize me! (SemEval-2022 Task 4 resources)," 2022. GitHub repository. [Online]. Available: `https://github.com/Perez-AlmendrosC/dontpatronizeme/tree/master/semeval-2022`

[3] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv preprint arXiv:1907.11692*, 2019. [Online]. Available: `https://arxiv.org/abs/1907.11692`

[4] M. Junczys-Dowmunt, R. Grundkiewicz, T. Dwojak, H. Hoang, K. Heafield, T. Neckermann, F. Seide, U. Germann, A. F. Aji, N. Bogoychev, A. F. T. Martins, and A. Birch, "Marian: Fast neural machine translation in C++," *arXiv preprint arXiv:1804.00344*, 2018. [Online]. Available: `https://arxiv.org/abs/1804.00344`

A Appendix: Detailed Figures

A.1 EDA 1 Detailed Figures

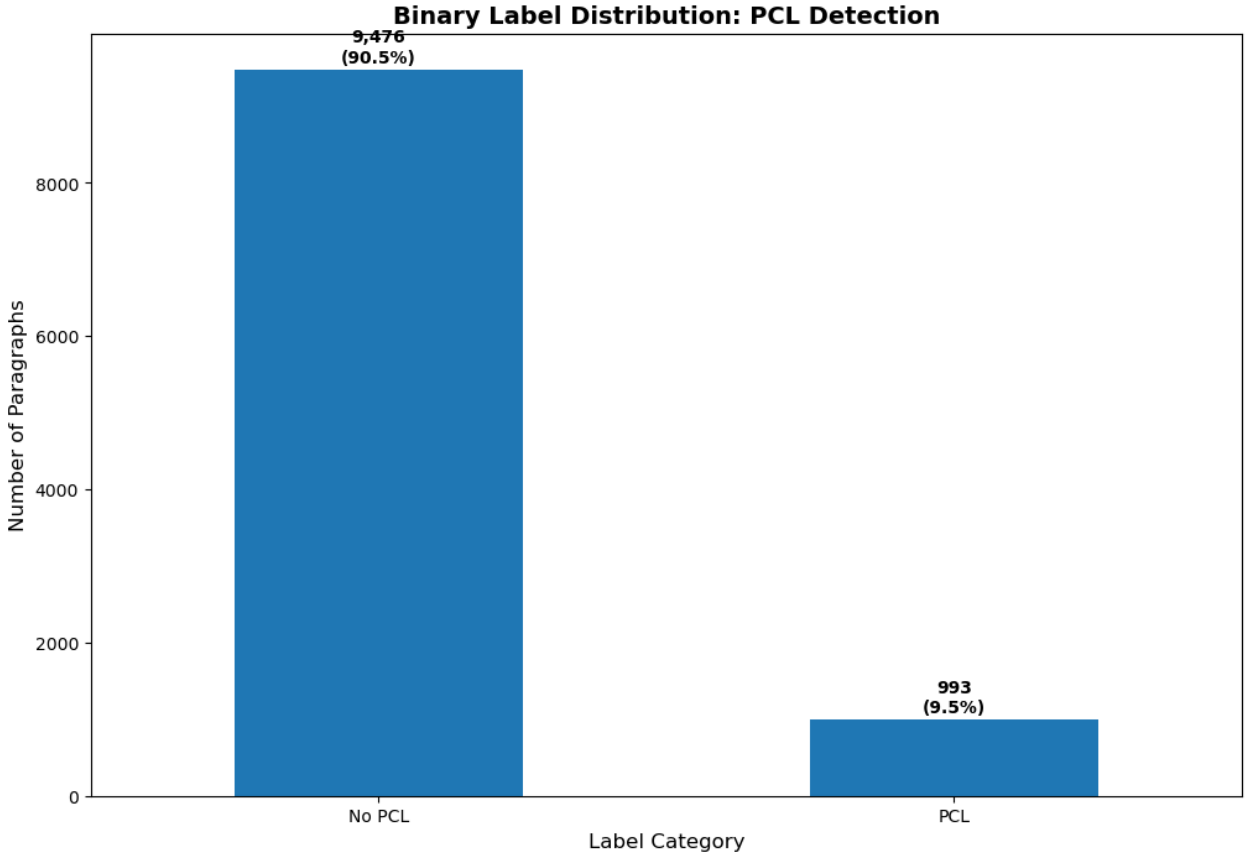


Figure 7: (Enlarged) Binary label distribution (No PCL vs PCL).

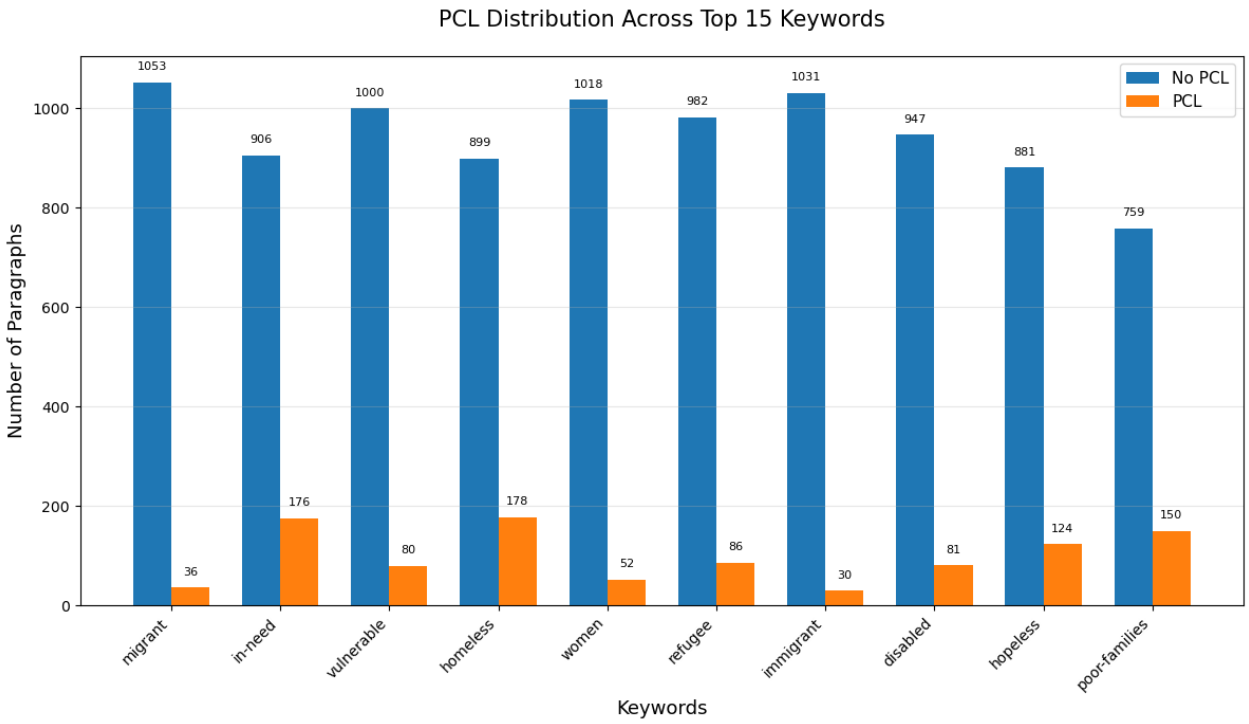


Figure 8: (Enlarged) PCL occurrence rate by Keywords (Top 15).

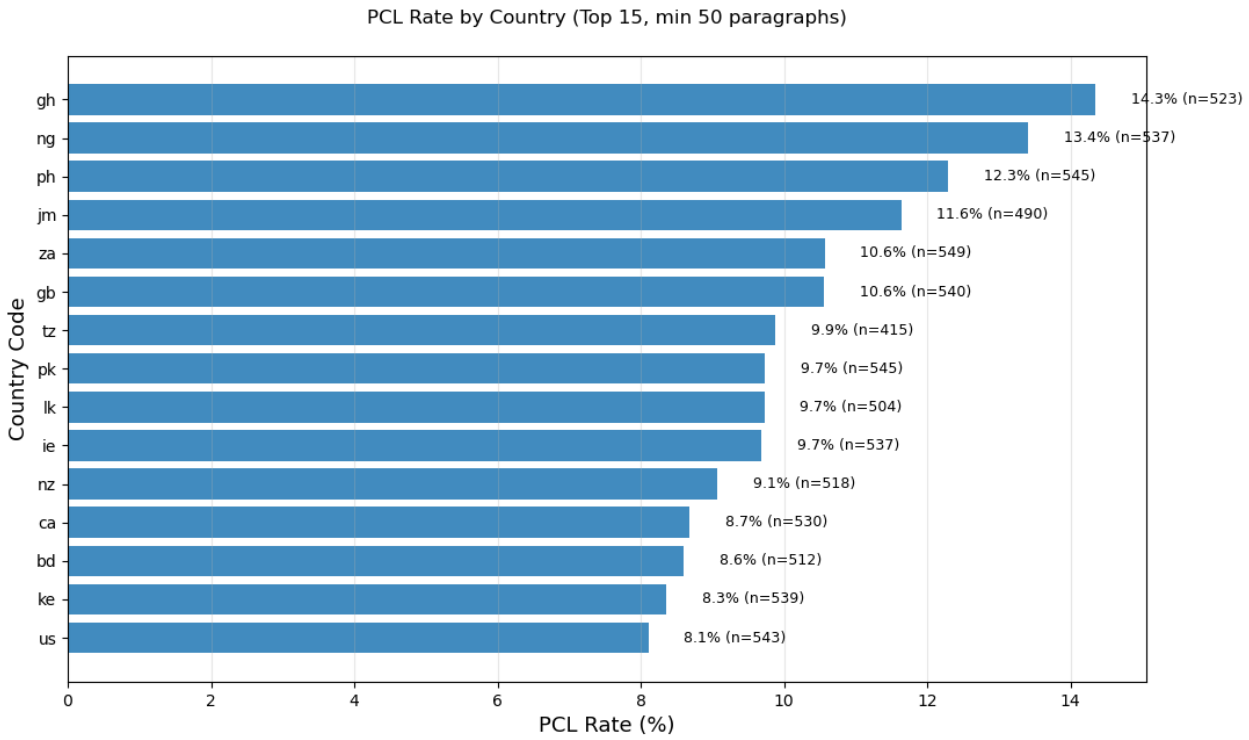


Figure 9: (Enlarged) PCL occurrence rate by Countries (Top 15).

A.2 EDA 2 Detailed Figures

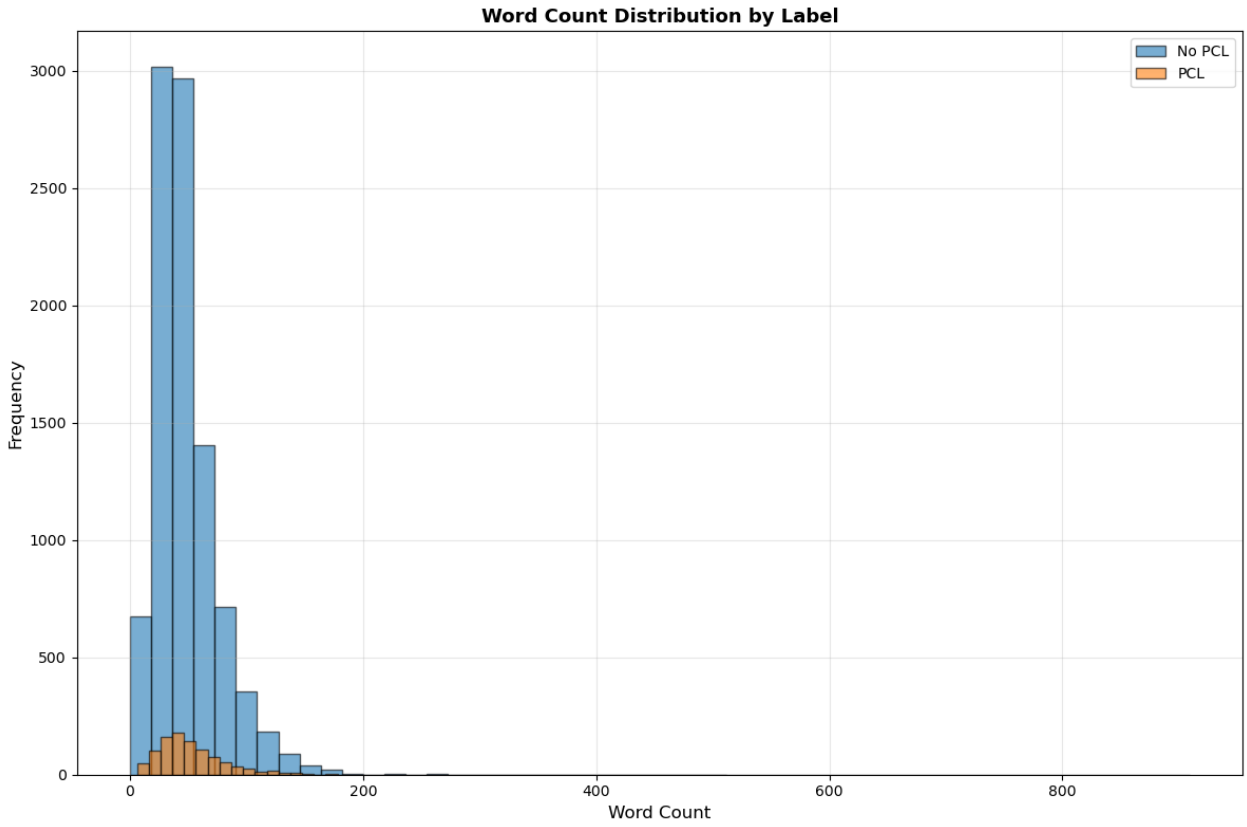


Figure 10: (Enlarged) Word count distribution by label.

B Appendix: Error Analysis Examples

Section counts

Section counts from the dev-set comparison were:

$$A = 1801, \quad B = 69, \quad C = 130, \quad D = 94.$$

A_both_correct (showing up to 5 examples)

- **idx=165** id=NA true=0 ours=0 base=0
  Text: Asylum seekers from Somalia were asked to report to Dadaab while those from other countries were asked to report to Kakuma refugee camp ...
- **idx=166** id=NA true=0 ours=0 base=0
  Text: Last week, the rights office reported dozens of execution-style killings in villages near Mosul, including the shooting of a physically disabled girl who failed to keep up on a forced march ...
- **idx=167** id=NA true=0 ours=0 base=0
  Text: Recommended changes to residential consumer tariff schemes will be designed to give more relief to poor families, especially those living in compound houses ...
- **idx=168** id=NA true=0 ours=0 base=0
  Text: Greek authorities say that 103 migrants picked up off a crippled yacht are being taken to port on the southern island of Crete ...
- **idx=170** id=NA true=0 ours=0 base=0
  Text: After 1896, Chinese immigrants had to have a guarantor and the poll tax had to be paid in advance. The tax was waived after 1934 but not repealed until 1944 ...

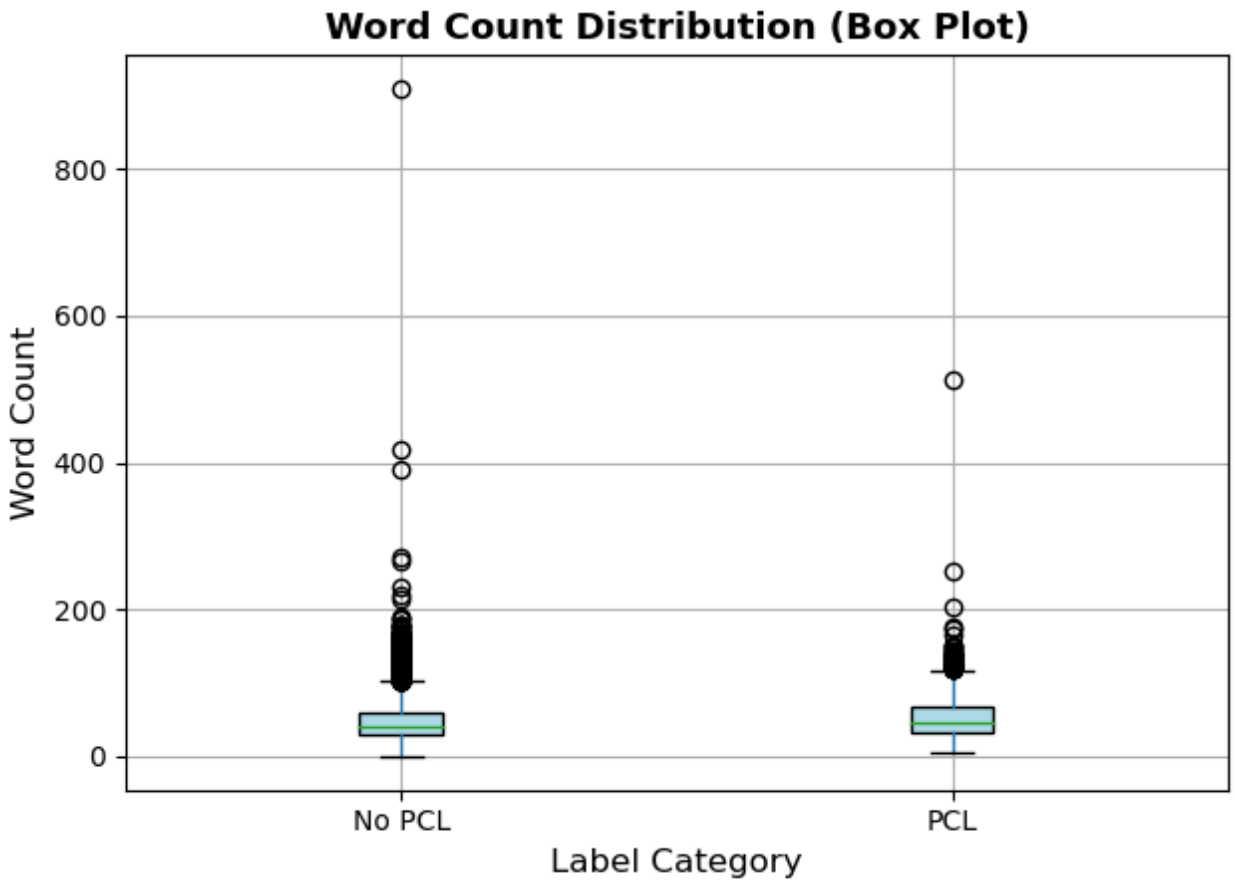


Figure 11: (Enlarged) Word count box plot by label.

B_both_wrong (showing up to 5 examples)

- **idx=0** id=NA true=1 ours=0 base=0
  Text: His present "chambers" may be quite humble, but Shiyani has the tiny space very neatly organized and clean ...
- **idx=1** id=NA true=1 ours=0 base=0
  Text: Krueger recently harnessed that creativity to self-publish a book featuring the poems, artwork, photography and short stories of 16 ill or disabled artists ...
- **idx=4** id=NA true=1 ours=0 base=0
  Text: We are alarmed to learn of your recently circulated proposals that would eviscerate the Lifeline program and leave many of the most vulnerable people in the country without access ...
- **idx=12** id=NA true=1 ours=0 base=0
  Text: It's calculated that over 204,000 days of purpose-built residential accommodation to otherwise potentially homeless elderly men and women have been delivered ...
- **idx=13** id=NA true=1 ours=0 base=0
  Text: "I and my daughter Monica are excited about providing a space for disabled people to be able to get together and earn fair prices for their work," Mr. Rogers said ...

C_ours_correct_base_wrong (showing up to 5 examples)

- **idx=2** id=NA true=1 ours=1 base=0
  Text: Parents of children who died must get compensation, free medicine must be provided to poor families across UP ...
- **idx=3** id=NA true=1 ours=1 base=0
  Text: You now see people without deformities begging, when some people who are disabled work to feed their mouth ...
- **idx=5** id=NA true=1 ours=1 base=0
  Text: "We share a global responsibility to respond to this crisis. We commend others who have acted in a compassionate and generous way ..."
- **idx=6** id=NA true=1 ours=1 base=0
  Text: The former Chelsea star through his foundation gave out toys, bags and clothes to kids in need of a brighter holiday ...
- **idx=7** id=NA true=1 ours=1 base=0
  Text: It can not be right to allow homes to sit empty while many struggle to find somewhere to live ... hoping against hope for some charity to provide shelter ...

D_base_correct_ours_wrong (showing up to 5 examples)

- **idx=169** id=NA true=0 ours=1 base=0
  Text: Marcos said the government should help poor families that try every possible means to survive ...
- **idx=252** id=NA true=0 ours=1 base=0
  Text: So, let the NPP government appointees look down upon the hardworking and selfless men and women at their own peril ...
- **idx=254** id=NA true=0 ours=1 base=0
  Text: "So we do need to heal ourselves as an Aboriginal Torres Strait islander community, but also as a nation" ...
- **idx=267** id=NA true=0 ours=1 base=0
  Text: "As a woman, I want to stand up for my reproductive rights ... As a human being, I want to stand up against Trump ..."
- **idx=287** id=NA true=0 ours=1 base=0
  Text: She added: "I would also like to carefully point out that the issue was not her religious beliefs, but rather it is about choosing to treat men and women differently ..."